COMP0222 Simultaneous Localisation and Mapping

Lecture 02: Using Kalman Filters for Linear SLAM

Simon Julier

s.julier@ucl.ac.uk

Sajad Saeedi

s.saeedi@ucl.ac.uk

Overview

- In the last couple of lectures, we've seen localization mapping and SLAM
- We've introduced some notation and description of the problem
- In this lecture, we will see how to implement SLAM using the Kalman Filter
- We will only consider a *linear* problem to separate the structure of the problem from the challenges of nonlinearities

Structure

- State Space Structure of the SLAM Problem
- Introduce the BeadSLAM Example
- Step-by-Step Example of the Kalman Filter SLAM System
- Generalizing Timesteps and Multiple Observations
- Analysing BeadSLAM
- Properties of Kalman Filter Based SLAM Systems

State Space Structure of the SLAM Problem

- **State Space Structure of the SLAM Problem**
- ***Introduce the BeadSLAM Example***
- ***Step-by-Step Example of the Kalman Filter SLAM System***
- ***Generalizing Timesteps and Multiple Observations***
- ***Analysing BeadSLAM***

Ground Truth State Space Model

- **This contains the robot state, and the state of all the landmarks,**

$$\mathbf{s}_k = \begin{pmatrix} \mathbf{x}_k \\ \mathbf{m} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_k \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{pmatrix}$$

Ground Truth State Space Model

Notation for Estimates

$$\hat{\mathbf{x}}_{i|j}, \ \mathbf{P}_{i|j}$$

Estimate at time i

Given observations up to time step j

When $i>j$ this is a *predicted* value
When $i=j$ this is a *filtered* value
When $i<j$ this is a *smoothed* value

The Kalman Estimate

- **The estimates are defined as**

$$\hat{\mathbf{x}}_{i|j} = \mathbb{E}\left[\mathbf{x}_i\right]$$

$$\mathbf{P}_{i|j} = \mathbb{E}\left[\tilde{\mathbf{x}}_{i|j}\tilde{\mathbf{x}}_{i|j}^{\top}\right]$$

Kalman Filter State

- **In the SLAM system, the estimate consists of the platform and the landmark estimates**

$$\hat{\mathbf{s}}_{k|k} = \begin{pmatrix} \hat{\mathbf{x}}_{k|k} \\ \hat{\mathbf{m}}_{k|k} \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{x}}_{k|k} \\ \hat{\mathbf{m}}^{1}_{k|k} \\ \vdots \\ \hat{\mathbf{m}}^{N_k}_{k|k} \end{pmatrix}$$

Kalman Filter State

Mapping Between Real-World and Filter Landmarks

- In general, the landmarks in the map differ from those in the filter:
  - The filter only maintains the landmarks its observed, not all the landmarks in the map
  - The is some assignment of real world landmarks to filter ones
- We simplify this by having the selector matrix O_k,

$$\mathrm{O}_k \mathrm{m}$$

Kalman Filter Covariance

- **The covariance is computed from**

$$\mathbf{P}^{s}_{k|k} = \mathbb{E}\left[\begin{pmatrix} \mathbf{x}_k - \hat{\mathbf{x}}_{k|k} \\ \mathbf{O}_k\mathbf{m} - \hat{\mathbf{m}}_{k|k} \end{pmatrix}\begin{pmatrix} \mathbf{x}_k - \hat{\mathbf{x}}_{k|k} \\ \mathbf{O}_k\mathbf{m} - \hat{\mathbf{m}}_{k|k} \end{pmatrix}^{\top}\right]$$

$$= \mathbb{E}\left[\begin{pmatrix} \tilde{\mathbf{x}}_{k|k} \\ \tilde{\mathbf{m}}_{k|k} \end{pmatrix}\begin{pmatrix} \tilde{\mathbf{x}}_{k|k} \\ \tilde{\mathbf{m}}_{k|k} \end{pmatrix}^{\top}\right]$$

The Structure of the Covariance Matrix

- **Taking expectations, the covariance matrix has the structure**

$$\mathbf{P}^{s}_{k|k} = \begin{bmatrix} \mathbf{P}^{\mathrm{xx}}_{k|k} & \mathbf{P}^{\mathrm{xm}}_{k|k} \\ \mathbf{P}^{\mathrm{mx}}_{k|k} & \mathbf{P}^{\mathrm{mm}}_{k|k} \end{bmatrix}$$

Platform Block

- The platform block is the covariance of the platform state,

$$\mathbf{P}^{\mathbf{xx}}_{k|k} = \mathbb{E}\left[\tilde{\mathbf{x}}_{k|k}\tilde{\mathbf{x}}^{\top}_{k|k}\right]$$

Landmark Block

- **This is the covariance of the landmark estimates**

$$\mathbf{P}^{\mathbf{mm}}_{k|k} = \mathbb{E}\left[\tilde{\mathbf{m}}_{k|k}\tilde{\mathbf{m}}^{\top}_{k|k}\right]$$

$$= \mathbb{E}\left[\begin{pmatrix}\tilde{\mathbf{m}}^{1}_{k|k}\\ \vdots \\ \tilde{\mathbf{m}}^{N_k}_{k|k}\end{pmatrix}\begin{pmatrix}\tilde{\mathbf{m}}^{1}_{k|k}\\ \vdots \\ \tilde{\mathbf{m}}^{N_k}_{k|k}\end{pmatrix}^{\top}\right]$$

Landmark Block

- **Expanding the landmark estimate terms,**

$$\mathbf{P}^{\mathbf{mm}}_{k|k} = \begin{bmatrix} \mathbf{P}^{11}_{k|k} & \mathbf{P}^{12}_{k|k} & \mathbf{P}^{13}_{k|k} & \cdots & \mathbf{P}^{1N_k}_{k|k} \\ \mathbf{P}^{21}_{k|k} & \mathbf{P}^{22}_{k|k} & \mathbf{P}^{23}_{k|k} & \cdots & \mathbf{P}^{2N_k}_{k|k} \\ \mathbf{P}^{31}_{k|k} & \mathbf{P}^{32}_{k|k} & \mathbf{P}^{33}_{k|k} & \cdots & \mathbf{P}^{3N_k}_{k|k} \\ \vdots & \vdots & \ddots & \vdots & \\ \mathbf{P}^{N_k1}_{k|k} & \mathbf{P}^{N_k2}_{k|k} & \mathbf{P}^{N_k3}_{k|k} & \cdots & \mathbf{P}^{N_kN_k}_{k|k} \end{bmatrix}$$

Platform-Landmark Block

- **This is the cross correlation between the vehicle and landmark estimates,**

$$\mathbf{P}^{\mathbf{xm}}_{k|k} = \mathbb{E}\left[\tilde{\mathbf{x}}_{k|k}\tilde{\mathbf{m}}^{\top}_{k|k}\right]$$

Platform-Landmark Block

- **Expanding the map error estimate,**

$$
\begin{aligned}
\mathbf{P}^{\mathbf{xm}}_{k|k} &= \mathbb{E}\left[\tilde{\mathbf{x}}_{k|k}\tilde{\mathbf{m}}^{\top}_{k|k}\right] \\
&= \mathbb{E}\left[\tilde{\mathbf{x}}_{k|k}\begin{pmatrix}\tilde{\mathbf{m}}^{1}_{k|k} & \cdots & \tilde{\mathbf{m}}^{N_k}_{k|k}\end{pmatrix}^{\top}\right] \\
&= \begin{bmatrix}\mathbf{P}^{\mathbf{x}1}_{k|k} & \cdots & \mathbf{P}^{\mathbf{x}N_k}_{k|k}\end{bmatrix}
\end{aligned}
$$

The Structure of the Covariance Matrix

- **Putting everything together,**

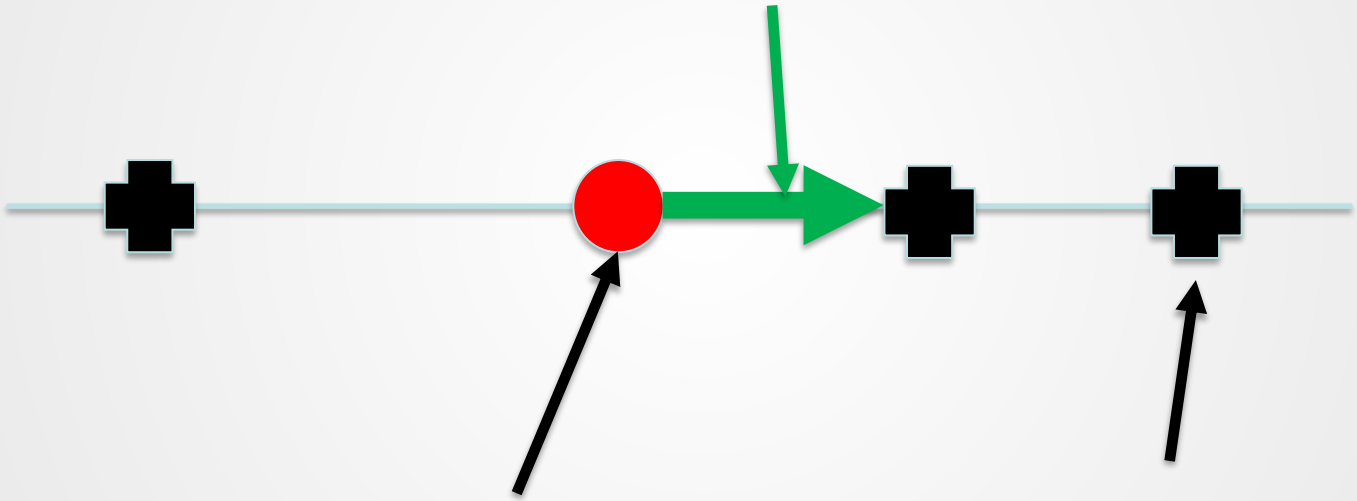
$$\mathbf{P}^{s}_{k|k} = \begin{bmatrix} \mathbf{P}^{\mathbf{xx}}_{k|k} & \mathbf{P}^{\mathbf{x}1}_{k|k} & \mathbf{P}^{\mathbf{x}2}_{k|k} & \mathbf{P}^{\mathbf{x}3}_{k|k} & \cdots & \mathbf{P}^{\mathbf{x}N_k}_{k|k} \\ \mathbf{P}^{1\mathbf{x}}_{k|k} & \mathbf{P}^{11}_{k|k} & \mathbf{P}^{12}_{k|k} & \mathbf{P}^{13}_{k|k} & \cdots & \mathbf{P}^{1N_k}_{k|k} \\ \mathbf{P}^{2\mathbf{x}}_{k|k} & \mathbf{P}^{21}_{k|k} & \mathbf{P}^{22}_{k|k} & \mathbf{P}^{23}_{k|k} & \cdots & \mathbf{P}^{2N_k}_{k|k} \\ \mathbf{P}^{3\mathbf{x}}_{k|k} & \mathbf{P}^{31}_{k|k} & \mathbf{P}^{32}_{k|k} & \mathbf{P}^{33}_{k|k} & \cdots & \mathbf{P}^{3N_k}_{k|k} \\ \vdots & \vdots & \ddots & \vdots & & \\ \mathbf{P}^{N_k\mathbf{x}}_{k|k} & \mathbf{P}^{N_k1}_{k|k} & \mathbf{P}^{N_k2}_{k|k} & \mathbf{P}^{N_k3}_{k|k} & \cdots & \mathbf{P}^{N_kN_k}_{k|k} \end{bmatrix}$$

Introduce the BeadSLAM Example

- *State Space Structure of the SLAM Problem*
- **Introduce the BeadSLAM Example**
- *Step-by-Step Example of the Kalman Filter SLAM System*
- *Generalizing Timesteps and Multiple Observations*
- *Analysing BeadSLAM*

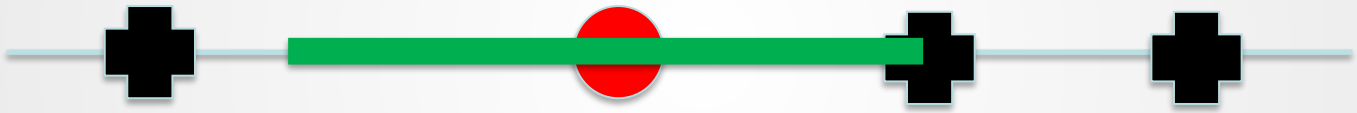
Simple Motivating Example… BeadSLAM

$$\mathbf{z}_k^i = [m^i - x_k]$$



$$\mathbf{x}_k = \begin{bmatrix} x_k \\ \dot{x}_k \end{bmatrix} \qquad \mathbf{m}^i = [m^i]$$

BeadSLAM Partial Observation of the Map



BeadSLAM Process Model

$$\mathbf{x}_{k+1} = \mathbf{F}_k \mathbf{x}_k + \mathbf{B}_k \mathbf{u}_k + \mathbf{v}_k$$

$$\mathbf{F}_k = \begin{bmatrix} 1 & \Delta T \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{B}_k = \begin{bmatrix} \Delta T^2/2 \\ \Delta T \end{bmatrix}$$

$$\mathbf{v}_k \sim \mathcal{G}\left(\mathbf{v}; \mathbf{0}, \mathbf{Q}\right)$$

$$\mathbf{Q}_k = \sigma_V^2 \begin{bmatrix} \Delta T^3/3 & \Delta T^2/2 \\ \Delta T^2/2 & \Delta T \end{bmatrix}$$

BeadSLAM Observation Model

$$\mathbf{z}_k^j = \mathbf{H}_k^{i_j} \mathbf{s}_k + \mathbf{w}_k^j$$

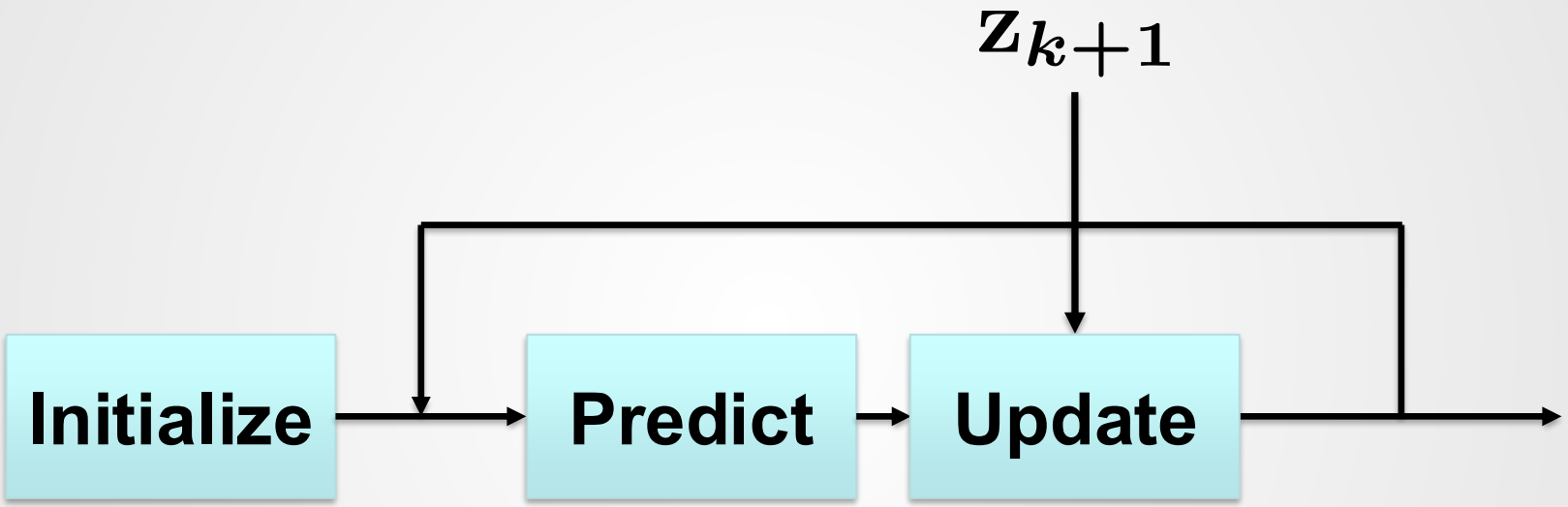
$$\mathbf{H}_k^{i_j} = \begin{bmatrix} -1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{bmatrix}$$

$$\mathbf{w}_k^j \sim \mathcal{G}\left(\mathbf{w}; \mathbf{0}, \mathbf{R}_k^j\right)$$

Step-by-Step Example of the Kalman Filter SLAM System

- *State Space Structure of the SLAM Problem*
- *Introduce the BeadSLAM Example*
- **Step-by-Step Example of the Kalman Filter SLAM System**
- *Generalizing Timesteps and Multiple Observations*
- *Analysing BeadSLAM*

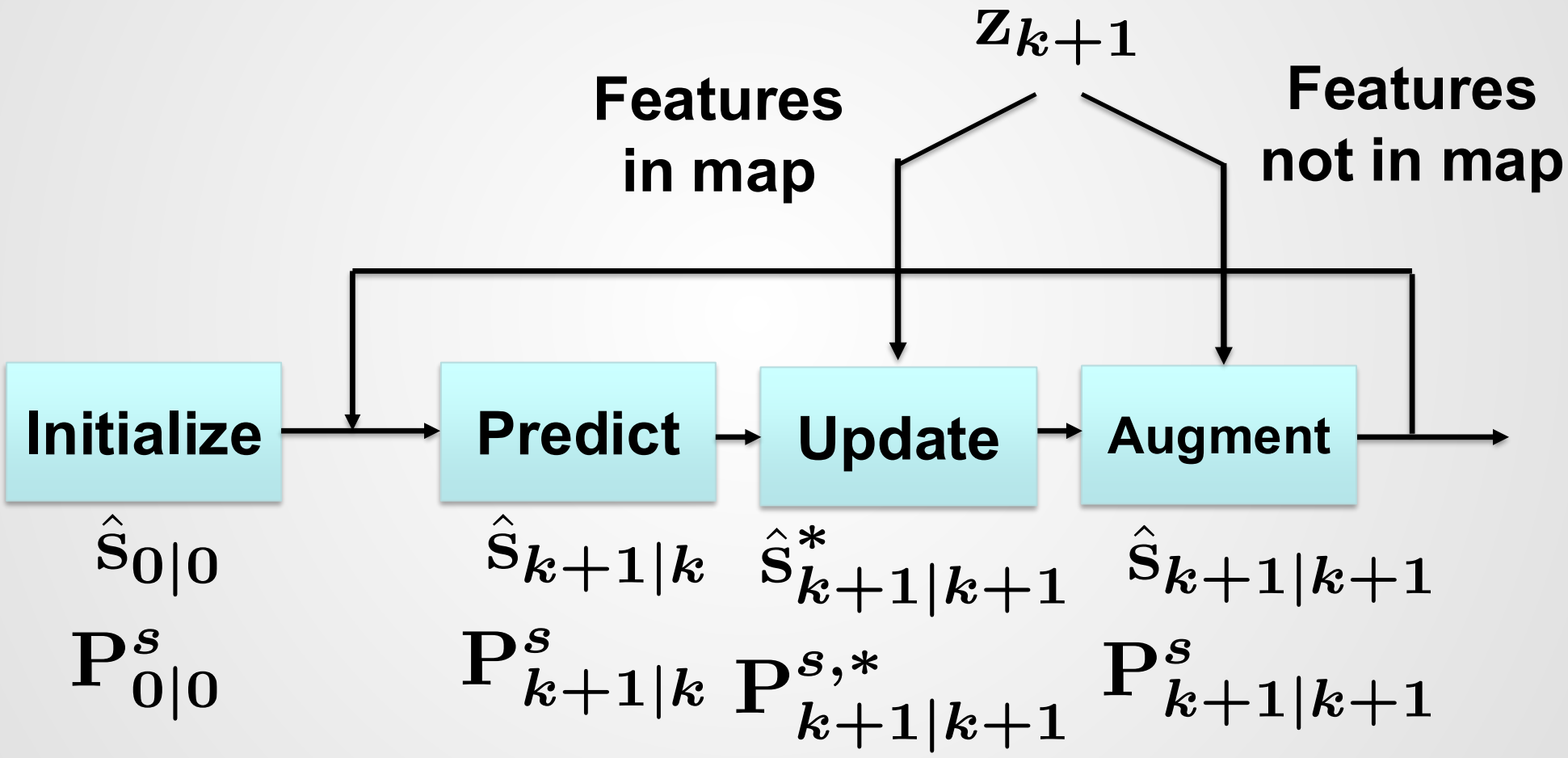
Standard Kalman Filter Structure



$\hat{\mathbf{s}}_{0|0}$ $\hat{\mathbf{s}}_{k+1|k}$ $\hat{\mathbf{s}}_{k+1|k+1}$

$\mathbf{P}^{s}_{0|0}$ $\mathbf{P}^{s}_{k+1|k}$ $\mathbf{P}^{s}_{k+1|k+1}$

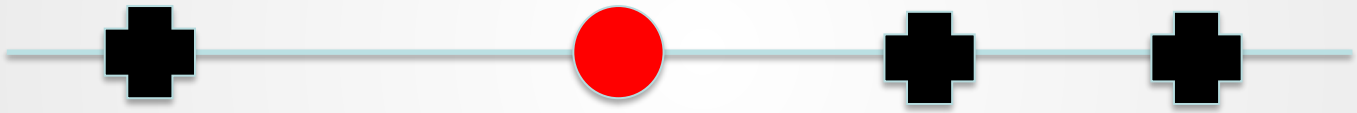
SLAM Kalman Filter Structure



BeadSLAM: The First Few Timesteps

Timestep	Action
0	Initialize
1	Predict and observe landmark 1
2	Predict and update landmark 1
3	Predict, observe landmark 2

Timestep 0: Initialize



Timestep 0: The Real World

- **The real world starts off with the platform at the origin and a fully populated map**

$$\mathbf{s}_0 = \begin{pmatrix} \mathbf{x}_0 \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{pmatrix}$$

Timestep 0: Initialization

Initialize

$$\hat{\mathbf{s}}_{0|0}$$

$$\mathbf{P}^{s}_{0|0}$$

Timestep 0: Initialise the Filter

- When the SLAM system starts, it has no information about landmarks
- The only information it has is the *platform state*,

$$\hat{\mathbf{s}}_{0|0} = \left[\hat{\mathbf{x}}_{0|0}\right]$$

$$\mathbf{P}^{\boldsymbol{s}}_{0|0} = \mathbb{E}\left[\tilde{\mathbf{x}}_{0|0}\tilde{\mathbf{x}}_{0|0}^{\top}\right]$$

Timestep 0: Initialise BeadSLAM

- The platform starts at the origin with no uncertainty
- Therefore,

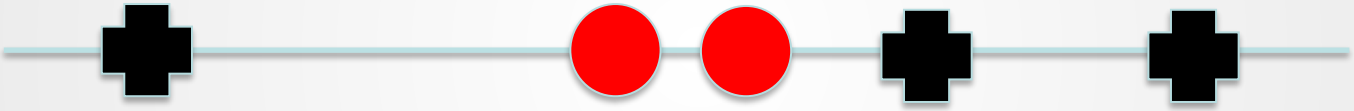
$$\hat{\mathbf{s}}_{0|0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\mathbf{P}^{s}_{0|0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

BeadSLAM: The First Few Timesteps

Timestep	Action
~~0~~	~~Initialize~~
1	Predict and observe landmark 1
2	Predict and update landmark 1
3	Predict, observe landmark 2

Timestep 1: Time Update in the True System



$$\mathbf{s}_0 \rightarrow \mathbf{s}_1$$

Timestep 1: Time Update in the True System

- **The platform state evolves according to the process model**

$$\mathbf{x}_{k+1} = \mathbf{F}_{k+1}\mathbf{x}_k + \mathbf{B}_{k+1}\mathbf{u}_{k+1} + \mathbf{v}_{k+1}$$

- **However, the landmarks *are stationary***
- **Therefore, their state *does not change* and *no process noise is injected into them***

Timestep 1: Time Update in the True System

- **Therefore, the real world state updates according to**

$$\mathbf{s}_1 = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{pmatrix} = \begin{pmatrix} \mathbf{F}_1\mathbf{x}_0 + \mathbf{B}_1\mathbf{u}_1 + \mathbf{v}_1 \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{pmatrix}$$

Timestep1: Filter Prediction

Initialize → **Predict**

$\hat{\mathbf{s}}_{0|0}$ $\hat{\mathbf{s}}_{1|0}$

$\mathbf{P}^{s}_{0|0}$ $\mathbf{P}^{s}_{1|0}$

Timestep 1: Filter Prediction

- **We want to compute**

$$\hat{\mathbf{s}}_{0|0} \rightarrow \hat{\mathbf{s}}_{1|0}$$

$$\mathbf{P}^{\boldsymbol{s}}_{0|0} \rightarrow \mathbf{P}^{\boldsymbol{s}}_{1|0}$$

Timestep 1: Filter Prediction

- **We need to apply the Kalman filter equations to the SLAM system state**

$$\hat{\mathbf{s}}_{1|0} = \mathbf{F}_1^{\mathbf{s}} \hat{\mathbf{s}}_{0|0} + \mathbf{B}_1^{\mathbf{s}} \mathbf{u}_1$$

$$\mathbf{P}_{1|0}^{\boldsymbol{s}} = \mathbf{F}_1^{\mathbf{s}} \mathbf{P}_{0|0}^{\boldsymbol{s}} \mathbf{F}_1^{\mathbf{s},\top} + \mathbf{Q}_1^{\boldsymbol{s}}$$

Timestep 1: Filter Prediction

- There are no landmarks, and so the SLAM state is the same as the platform state
- Therefore, we can directly use the platform process model,

$$\hat{\mathbf{s}}_{1|0} = \mathbf{F}_1 \hat{\mathbf{s}}_{0|0} + \mathbf{B}_1^{\mathbf{s}} \mathbf{u}_1$$

$$\mathbf{P}_{1|0}^{\boldsymbol{s}} = \mathbf{F}_1 \mathbf{P}_{0|0}^{\boldsymbol{s}} \mathbf{F}_1^{\mathbf{s},\top} + \mathbf{Q}_1^{\boldsymbol{s}}$$

Timestep 1: Observation Generation

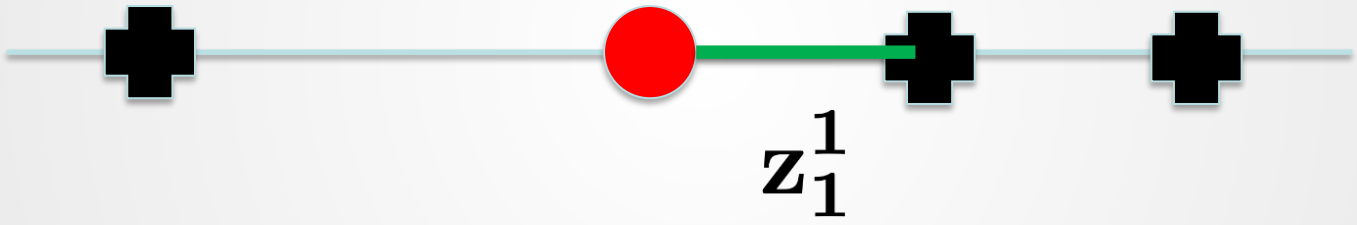
$$\mathbf{z}_1^1$$

Initialize → **Predict**

$$\hat{\mathbf{s}}_{0|0} \qquad \hat{\mathbf{s}}_{1|0}$$

$$\mathbf{P}^{\boldsymbol{s}}_{0|0} \qquad \mathbf{P}^{\boldsymbol{s}}_{1|0}$$

Timestep 1: Observation from True System



Timestep 1: Observation Generation

- We observe the landmark according to the observation model

$$\mathbf{z}_k^j = \mathbf{H}_k^{i_j} \mathbf{x}_k + \mathbf{w}_k^j$$

- For the BeadSLAM example,

$$\mathbf{z}_1^1 = \left[m^1 - x_1 + w_1^1 \right]$$

Timestep 1: Update with Known Landmarks

$\mathbf{z}_1^1$

Features in map

Initialize → **Predict** → **Update**

$\hat{\mathbf{s}}_{0|0}$ $\quad$ $\hat{\mathbf{s}}_{1|0}$ $\quad$ $\hat{\mathbf{s}}^*_{1|1}$

$\mathbf{P}^s_{0|0}$ $\quad$ $\mathbf{P}^s_{1|0}$ $\quad$ $\mathbf{P}^{s,*}_{1|1}$

Timestep 1: Update with Known Landmarks

- **The first part of the update step is to update the filter with observations of *known* landmarks,**

$$\hat{\mathbf{s}}_{1|0} \rightarrow \hat{\mathbf{s}}^{*}_{1|1}$$

$$\mathbf{P}^{s}_{1|0} \rightarrow \mathbf{P}^{s,*}_{1|1}$$

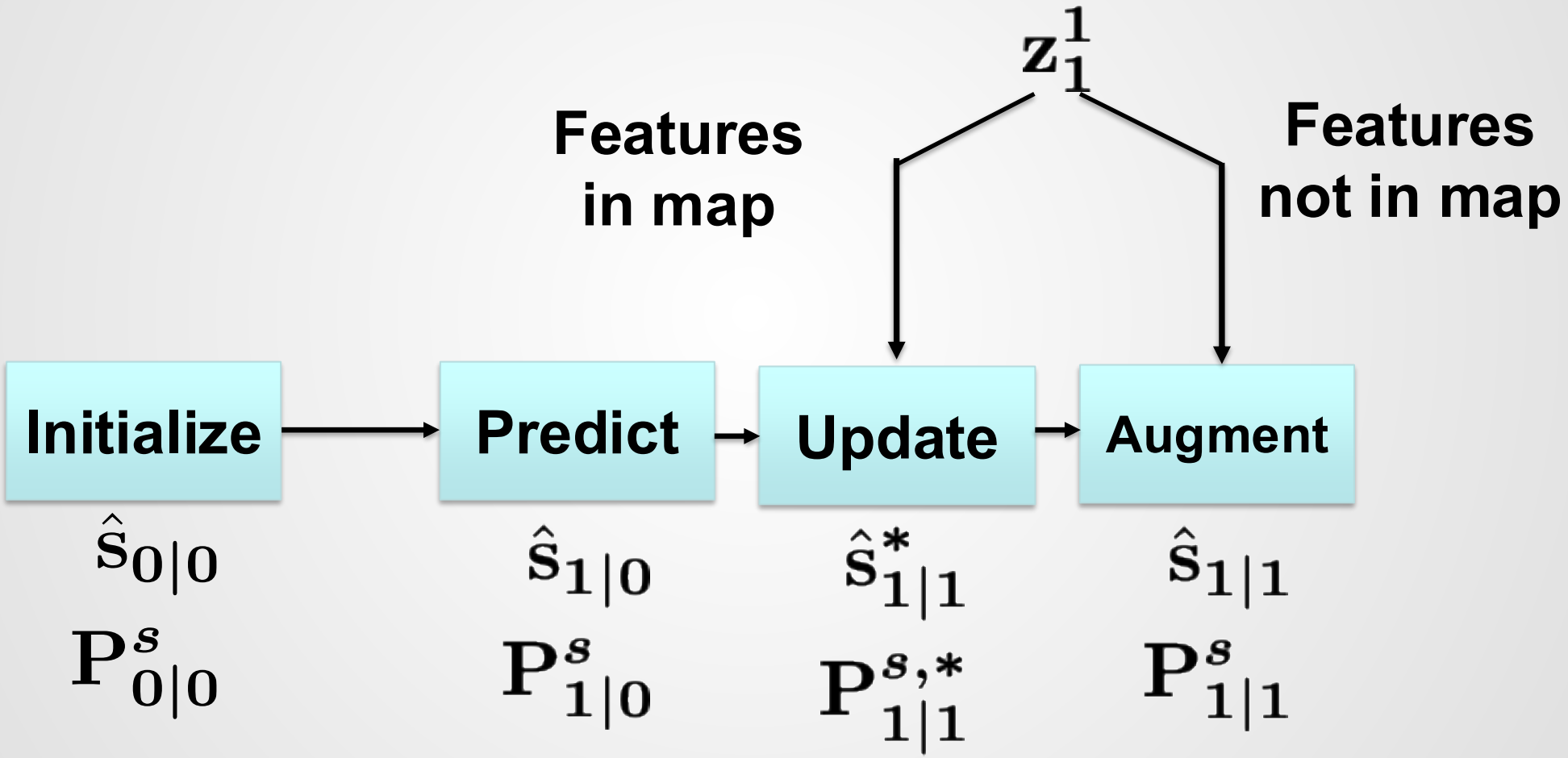
Timestep 1: Update with Known Landmarks

- Since the map is empty at time step 1, we have no existing landmarks
- Therefore, this operation is a no-op and

$$\hat{\mathbf{s}}^{*}_{1|1} = \hat{\mathbf{s}}_{1|0}$$

$$\mathbf{P}^{s,*}_{1|1} = \mathbf{P}^{s}_{1|0}$$

Timestep 1: Augment with New Landmarks



Timestep 1: Augment with New Landmarks

- **This step handles adding any new landmarks to the SLAM system,**

$$\hat{\mathbf{s}}^{*}_{1|1} \rightarrow \hat{\mathbf{s}}_{1|1}$$

$$\mathbf{P}^{s,*}_{1|1} \rightarrow \mathbf{P}^{s}_{1|1}$$

- **This must *increase* the dimension of the state space**

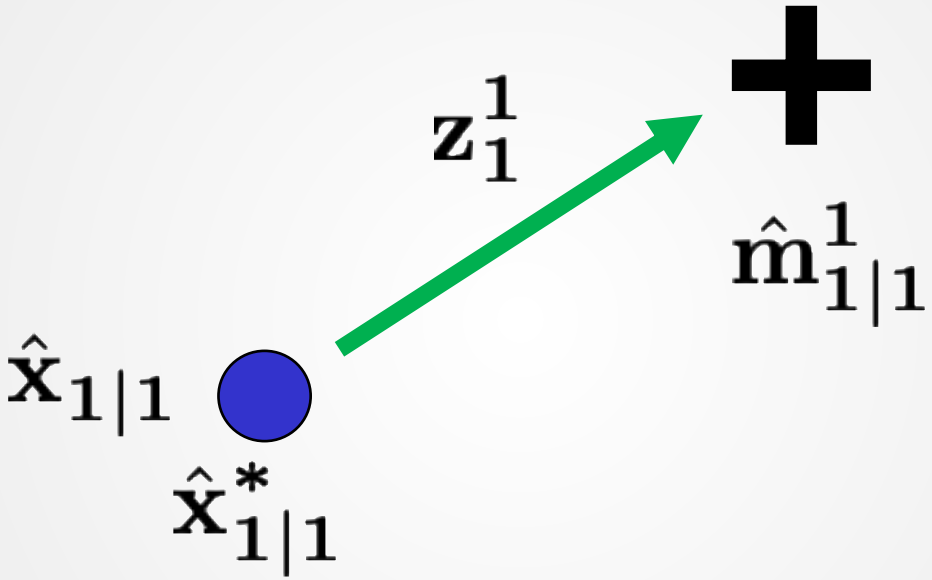
Timestep 1: Augment with New Landmarks

- It's easiest to start by thinking about what happens with the mean
- The augmentation operation transforms the state vector

$$\hat{\mathbf{s}}^{*}_{1|1} \rightarrow \hat{\mathbf{s}}_{1|1}$$

$$\begin{pmatrix} \hat{\mathbf{x}}^{*}_{1|1} \end{pmatrix} \rightarrow \begin{pmatrix} \hat{\mathbf{x}}_{1|1} \\ \hat{\mathbf{m}}^{1}_{1|1} \end{pmatrix}$$

Timestep 1: Augment with New Landmarks



Timestep 1: Augment with New Landmarks

- **We assume that initializing with a new landmark does not change the state of the platform estimate,**

$$\left(\hat{\mathbf{x}}^{*}_{1|1}\right) \rightarrow \begin{pmatrix} \hat{\mathbf{x}}^{*}_{1|1} \\ \hat{\mathbf{m}}^{1}_{1|1} \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{x}}_{1|1} \\ \hat{\mathbf{m}}^{1}_{1|1} \end{pmatrix}$$

Timestep 1: Augment with New Landmarks in BeadSLAM

- In BeadSLAM, we can figure out how to do this directly from the algebra
- Recall the observation model is

$$\mathbf{z}_1^1 = [z_1^1] = [m^1 - x_1 + w_1^1]$$

Timestep 1: Augment with New Landmarks in BeadSLAM

- **Therefore, we can rewrite this to compute the landmark position from the platform estimate and the observation,**

$$m^1 = z_1^1 + x_1 - w_1^1$$

Timestep 1: Augment with New Landmarks in BeadSLAM

- **Therefore, for BeadSLAM,**

$$\hat{\mathbf{s}}_{1|1} = \begin{pmatrix} \hat{x}_{1|1} \\ \hat{\dot{x}}_{1|1} \\ \hat{m}_{1|1} \end{pmatrix} = \begin{pmatrix} \hat{x}_{1|1} \\ \hat{\dot{x}}_{1|1} \\ \hat{x}_{1|1} + z_1^1 \end{pmatrix}$$

Timestep 1: Augmentation Transformation

- We can initialize the landmark mean from the inverse observation model
- However, we have to work out the covariance as well
- The way we do this is to explicitly write out the augmentation step as a *transformation* we apply to the Kalman filter estimate
- We can then compute the covariance from that transformation

Timestep 1: Augmentation Transformation

- **Since we are still dealing with linear process and observation models, the augmentation operation can be written in the form**

$$\hat{\mathbf{s}}_{1|1} = \mathbf{J}_1^{\mathbf{s}} \hat{\mathbf{s}}_{1|1}^{*} + \mathbf{K}_1^{\mathbf{s}} \mathbf{z}_1^1$$

$$\mathbf{J}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \ \mathbf{K}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Timestep 1: Augmentation Transformation

Timestep 1: Augmentation Transformation

- Given this linear transformation, the covariance of the augmented Kalman filter is

$$\mathbf{P}^{s}_{k+1|k+1} = \mathbf{J}_1 \mathbf{P}^{s,*}_{1|1} \mathbf{J}_1^\top + \mathbf{K}_1 \mathbf{R}^1_1 \mathbf{K}_1$$

Timestep 1: Augmentation Transformation

- **Before augmentation, the covariance has the structure**

$$\mathbf{P}^{s}_{1|1} = \begin{bmatrix} \mathbf{P}^{\mathbf{xx}}_{1|1} & \mathbf{P}^{\mathbf{x1}}_{1|1} \\ \mathbf{P}^{\mathbf{1x}}_{1|1} & \mathbf{P}^{\mathbf{11}}_{1|1} \end{bmatrix}$$

Timestep 1: Augmented Covariance

$$\mathbf{P}^{s}_{1|1} = \mathbf{J}_1 \mathbf{P}^{s,*}_{1|1} \mathbf{J}_1^\top + \mathbf{K}_1 \mathbf{R}^1_1 \mathbf{K}_1^\top$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\dot{\boldsymbol{x}}^*} \\ \boldsymbol{P}^{\dot{\boldsymbol{x}}^*\boldsymbol{x}^*} & \boldsymbol{P}^{\dot{\boldsymbol{x}}^*\dot{\boldsymbol{x}}^*} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}^\top$$

$$+ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{R}^1_1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}^\top$$

Timestep 1: Augmented Covariance

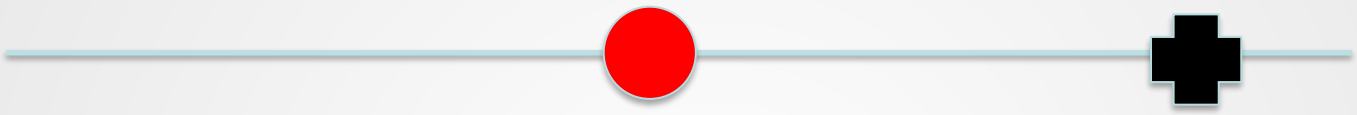
$$\mathbf{P}^{s}_{1|1} = \mathbf{J}_1 \mathbf{P}^{s,*}_{1|1} \mathbf{J}_1^\top + \mathbf{K}_1 \mathbf{R}^1_1 \mathbf{K}_1^\top$$

$$= \begin{bmatrix} \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\dot{\boldsymbol{x}}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} \\ \boldsymbol{P}^{\dot{\boldsymbol{x}}^*\boldsymbol{x}^*} & \boldsymbol{P}^{\dot{\boldsymbol{x}}^*\dot{\boldsymbol{x}}^*} & \boldsymbol{P}^{\dot{\boldsymbol{x}}^*\boldsymbol{x}^*} \\ \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\dot{\boldsymbol{x}}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} + \boldsymbol{R}^1_1 \end{bmatrix}$$

Interpreting the Augmented Covariance

- Interpreting what's going on with this covariance structure is *absolutely fundamental* to what's going on with SLAM
- Therefore, we are going to spend a few minutes thinking a bit more about what's happening
- We will consider three cases:
  1. The platform state is known *perfectly*, and the sensors is *noisy*
  2. The platform state is known *imperfectly*, and the sensor is *noiseless*
  3. The platform state is known *imperfectly*, and the sensors is *noisy*

Case 1: Platform State Perfect, Sensor Noisy

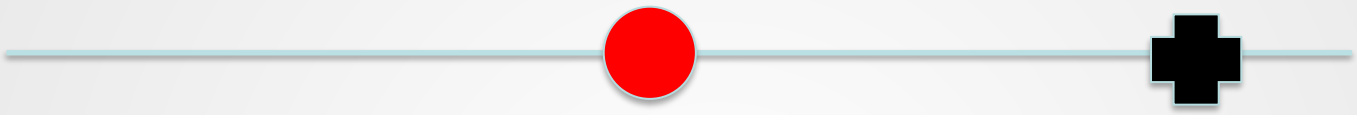


Case 1: Platform State Perfect, Sensor Noisy

$$\mathbf{P}^{s}_{1|1} = \mathbf{J}_1 \mathbf{P}^{s,*}_{1|1} \mathbf{J}_1^\top + \mathbf{K}_1 \mathbf{R}^1_1 \mathbf{K}_1^\top$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & R^1_1 \end{bmatrix}$$

Case 2: Platform State Imperfect, Sensor Noiseless

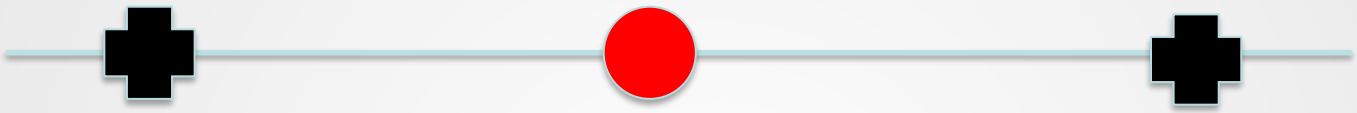


Case 2: Platform State Imperfect, Sensor Noiseless

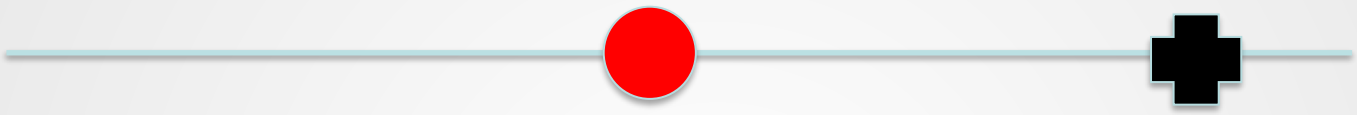
$$\mathbf{P}^{s}_{1|1} = \mathbf{J}_1 \mathbf{P}^{s,*}_{1|1} \mathbf{J}_1^\top + \mathbf{K}_1 \mathbf{R}^1_1 \mathbf{K}_1^\top$$

$$= \begin{bmatrix} P^{x^* x^*} & P^{x^* \dot{x}^*} & P^{x^* x^*} \\ P^{\dot{x}^* x^*} & P^{\dot{x}^* \dot{x}^*} & P^{\dot{x}^* x^*} \\ P^{x^* x^*} & P^{x^* \dot{x}^*} & P^{x^* x^*} \end{bmatrix}$$

Case 2: Platform State Imperfect, Sensor Noiseless



Case 3: Platform State Imperfect, Sensor Noisy



Case 3: Platform State Imperfect, Sensor Noisy

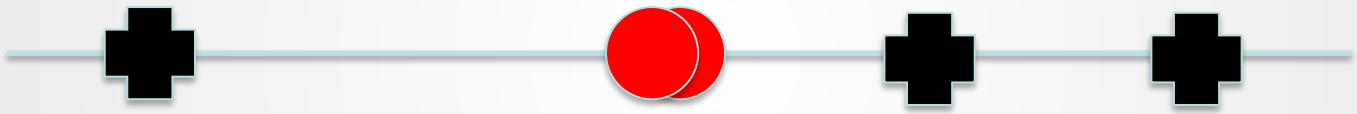
$$\mathbf{P}^{s}_{1|1} = \mathbf{J}_1 \mathbf{P}^{s,*}_{1|1} \mathbf{J}_1^\top + \mathbf{K}_1 \mathbf{R}^1_1 \mathbf{K}_1^\top$$

$$= \begin{bmatrix} \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\dot{\boldsymbol{x}}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} \\ \boldsymbol{P}^{\dot{\boldsymbol{x}}^*\boldsymbol{x}^*} & \boldsymbol{P}^{\dot{\boldsymbol{x}}^*\dot{\boldsymbol{x}}^*} & \boldsymbol{P}^{\dot{\boldsymbol{x}}^*\boldsymbol{x}^*} \\ \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\dot{\boldsymbol{x}}^*} & \boldsymbol{P}^{\boldsymbol{x}^*\boldsymbol{x}^*} + \boldsymbol{R}^1_1 \end{bmatrix}$$

BeadSLAM: The First Few Timesteps

Timestep	Action
~~0~~	~~Initialize~~
~~1~~	~~Predict and observe landmark 1~~
2	Predict and update landmark 1
3	Predict, observe landmark 2

Timestep 2: True System Time Update



Timestep 2: Time Update in the True System

- **The true state updates the same way as before,**

$$\mathbf{s}_2 = \begin{bmatrix} \mathbf{x}_2 \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{bmatrix} = \begin{bmatrix} \mathbf{F}_1\mathbf{x}_1 + \mathbf{B}_1\mathbf{u}_1 + \mathbf{v}_1 \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{bmatrix}$$

Timestep 2: Prediction

Predict

$$\hat{\mathbf{s}}_{2|1}$$

$$\mathbf{P}^{s}_{2|1}$$

Timestep 2: Kalman Filter Prediction

- **We predict the Kalman Filter using,**

$$\hat{\mathbf{s}}_{2|1} = \mathbf{F}_1^{\mathbf{s}} \hat{\mathbf{s}}_{1|1} + \mathbf{B}_1^{\mathbf{s}} \mathbf{u}_1$$

$$\mathbf{P}_{2|1}^{s} = \mathbf{F}_1^{\mathbf{s}} \mathbf{P}_{1|1}^{s} \mathbf{F}_1^{\mathbf{s}\top} + \mathbf{Q}_1^{s}$$

Timestep 2: Kalman Filter Prediction

- We have to determine the structure of the process model which will work with the SLAM state
- However, we only have the process model for the platform state
- Therefore, we have to synthesize a process model from the platform model
- We use the behaviour of the true system to guide this

Timestep 2: True System Redux

- **Recall the prediction has the form**

$$\mathbf{s}_2 = \begin{bmatrix} \mathbf{x}_2 \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{bmatrix} = \begin{bmatrix} \mathbf{F}_1\mathbf{x}_1 + \mathbf{B}_1\mathbf{u}_1 + \mathbf{v}_1 \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{bmatrix}$$

- **The landmark states do not change and no process noise is added to them**

Timestep 2: Kalman Filter Prediction

- **Therefore, we set up our process model so that the landmark states and covariances do not change,**

$$\mathbf{F}_1^s = \begin{bmatrix} \mathbf{F}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}, \ \mathbf{B}_1^s = \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{0} \end{bmatrix}, \ \mathbf{Q}_1^s = \begin{bmatrix} \mathbf{Q}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$$

Timestep 2: Kalman Filter Mean Prediction

- **Substituting, the estimate updates according to**

$$\hat{\mathbf{s}}_{2|1} = \begin{bmatrix} \mathbf{F}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}}_{1|1} \\ \hat{\mathbf{m}}^1_{1|1} \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{0} \end{bmatrix} \mathbf{u}_1$$

$$= \begin{bmatrix} \mathbf{F}_1 \hat{\mathbf{x}}_{1|1} + \mathbf{B}_1 \mathbf{u}_1 \\ \hat{\mathbf{m}}^1_{1|1} \end{bmatrix}$$

Timestep 2: Kalman Filter Covariance Prediction

- The covariance is predicted from

$$\mathbf{P}^{s}_{2|1} = \begin{bmatrix} \mathbf{F}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{P}^{\mathbf{xx}}_{1|1} & \mathbf{P}^{\mathbf{x1}}_{1|1} \\ \mathbf{P}^{\mathbf{1x}}_{1|1} & \mathbf{P}^{\mathbf{11}}_{1|1} \end{bmatrix} \begin{bmatrix} \mathbf{F}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}^{\top} + \begin{bmatrix} \mathbf{Q}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}$$

Timestep 2: Kalman Filter Covariance Prediction

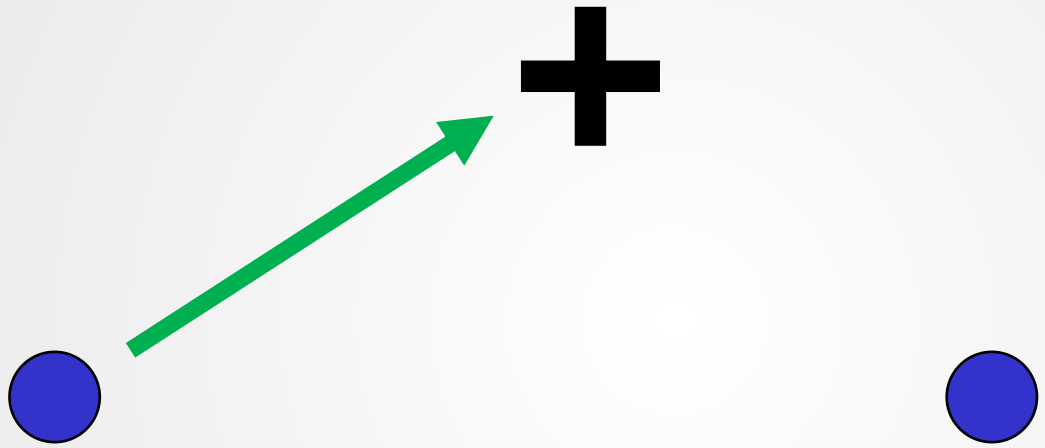
- **Expanding,**

$$\mathbf{P}^{s}_{2|1} = \begin{bmatrix} \mathbf{F}_1 \mathbf{P}^{\mathbf{xx}}_{1|1} \mathbf{F}_1^\top + \mathbf{Q}_1 & \mathbf{F}_1 \mathbf{P}^{\mathbf{x1}}_{1|1} \\ \mathbf{P}^{\mathbf{1x}}_{1|1} \mathbf{F}_1^\top & \mathbf{P}^{\mathbf{11}}_{1|1} \end{bmatrix}$$

Timestep 2: Interpreting the Prediction Covariance

$$\begin{bmatrix} \mathbf{F}_1 \mathbf{P}^{\mathbf{xx}}_{1|1} \mathbf{F}_1^\top + \mathbf{Q}_1 & \mathbf{F}_1 \mathbf{P}^{\mathbf{x1}}_{1|1} \\ \mathbf{P}^{\mathbf{1x}}_{1|1} \mathbf{F}_1^\top & \mathbf{P}^{\mathbf{11}}_{1|1} \end{bmatrix}$$

Timestep 2: Interpreting the Prediction Covariance



Timestep 2: Expanding the Cross Correlation

- We can see the effect if we look at BeadSLAM
- Recall that

$$\mathbf{P}^{s}_{1|1} = \begin{bmatrix} P^{x^*x^*} & P^{x^*\dot{x}^*} & P^{x^*x^*} \\ P^{\dot{x}^*x^*} & P^{\dot{x}^*\dot{x}^*} & P^{\dot{x}^*x^*} \\ P^{x^*x^*} & P^{x^*\dot{x}^*} & P^{x^*x^*} + R^1_1 \end{bmatrix}$$

Timestep 2: Expanding the Cross Correlation

- **Substituting,**

$$\mathbf{F}_1\mathbf{P}_{1|1}^{\mathbf{x}1} = \begin{bmatrix} 1 & \Delta T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} P^{x^*x^*} \\ P^{\dot{x}^*x^*} \end{bmatrix}$$

$$= \begin{bmatrix} P^{x^*x^*} + \Delta T P^{\dot{x}^*x^*} \\ P^{\dot{x}^*x^*} \end{bmatrix}$$

Timestep 2: Observe Landmark 1

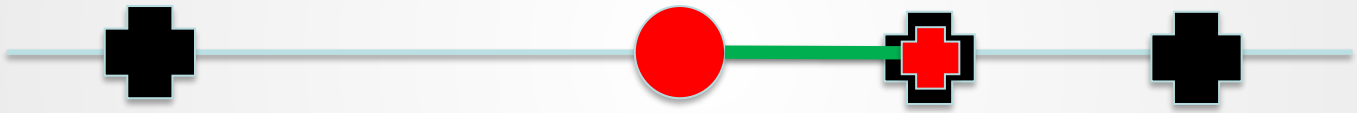
$$\mathbf{z}_2^1$$

Predict

$$\hat{\mathbf{s}}_{2|1}$$

$$\mathbf{P}_{2|1}^{s}$$

Timestep 2: Observe Landmark 1



Timestep 2: Update with Known Landmarks

$\mathbf{z}_2^1$

Features in map

Predict → **Update**

$\hat{\mathbf{s}}_{2|1}$ $\hat{\mathbf{s}}^*_{2|2}$

$\mathbf{P}^s_{2|1}$ $\mathbf{P}^{s,*}_{2|2}$

Timestep 2: Update with Known Landmarks

- **We update using the standard Kalman filter equations**

$$\hat{\mathbf{s}}^{*}_{2|2} = \hat{\mathbf{s}}_{2|1} + \mathbf{W}_{2|1}\boldsymbol{\nu}_{2|1}$$

$$\mathbf{P}^{s,*}_{2|2} = \mathbf{P}^{s}_{2|1} - \mathbf{W}_{2|1}\mathbf{S}_{2|1}\mathbf{W}^{\top}_{2|1}$$

Timestep 2: Update with Known Landmarks

$$\boldsymbol{\nu}_{2|1} = \mathbf{z}_2^1 - \mathbf{H}_2^1 \hat{\mathbf{s}}_{2|1}$$

$$\mathbf{C}_{2|1} = \mathbf{P}_{2|1}^{s} \mathbf{H}_2^{s\top}$$

$$\mathbf{S}_{2|1} = \mathbf{H}_2^1 \mathbf{C}_{2|1} + \mathbf{R}_2^1$$

$$\mathbf{W}_{2|1} = \mathbf{C}_{2|1} \mathbf{S}_{2|1}^{-1}$$

Timestep 2: Properties of the Update Step

- **The covariance does *not increase* during an update**

- **The states which are updated *are controlled by the structure* of $\mathbf{C}_{2|1}$**

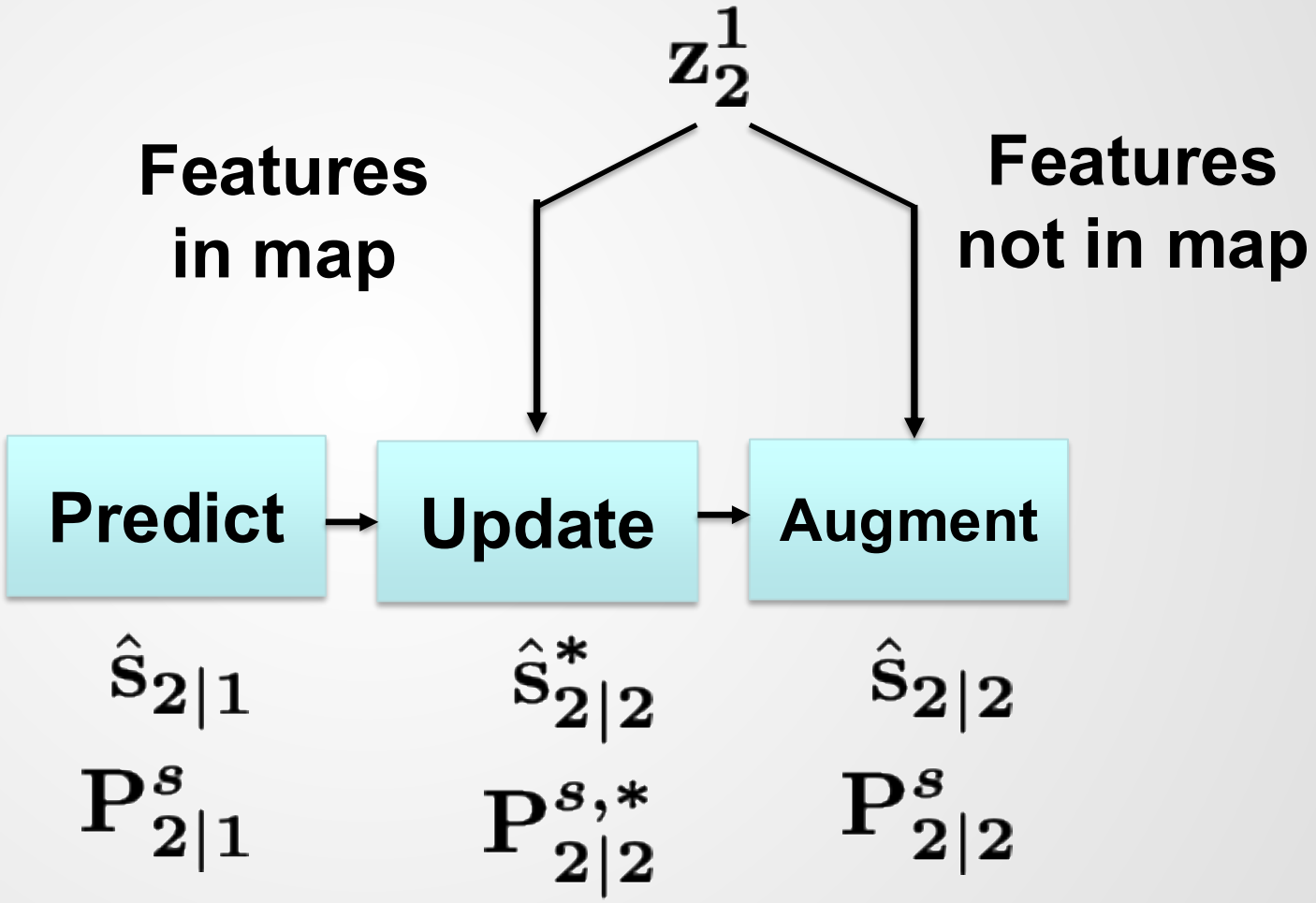
Timestep 2: Gain Matrix Structure

$$\mathbf{C}_{2|1} = \mathbf{P}^{s}_{2|1}\mathbf{H}^{s\top}_{2}$$

$$= \begin{bmatrix} P^{xx} & P^{x\dot{x}} & P^{xm^1} \\ P^{\dot{x}x} & P^{\dot{x}\dot{x}} & P^{\dot{x}m^1} \\ P^{m^1x} & P^{m^1\dot{x}} & P^{m^1m^1} \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} P^{xm^1} - P^{xx} \\ P^{\dot{x}m^1} - P^{\dot{x}x} \\ P^{m^1m^1} - P^{m^1x} \end{bmatrix}$$

Timestep 2: Augment with New Landmarks



Timestep 2: Augment with New Landmarks

- Since we only observe landmark 1, this is already in the map
- Therefore, augment is a no-op and so

$$\hat{\mathbf{s}}_{2|2} = \hat{\mathbf{s}}^{*}_{2|2}$$

$$\mathbf{P}^{s}_{2|2} = \mathbf{P}^{s,*}_{2|2}$$

BeadSLAM: The First Few Timesteps

Timestep	Action
~~0~~	~~Initialize~~
~~1~~	~~Predict and observe landmark 1~~
~~2~~	~~Predict and update landmark 1~~
3	Predict, observe landmark 2

Timestep 3: Predict and Observe Landmark 2

Timestep 3: Predict and Observe Landmark 2

- The predict step is the same as before
- We do not observe landmark 1 and so the predicted estimate is the partially updated estimate
- However, we now need to see what happens when we augment a map with a second landmark

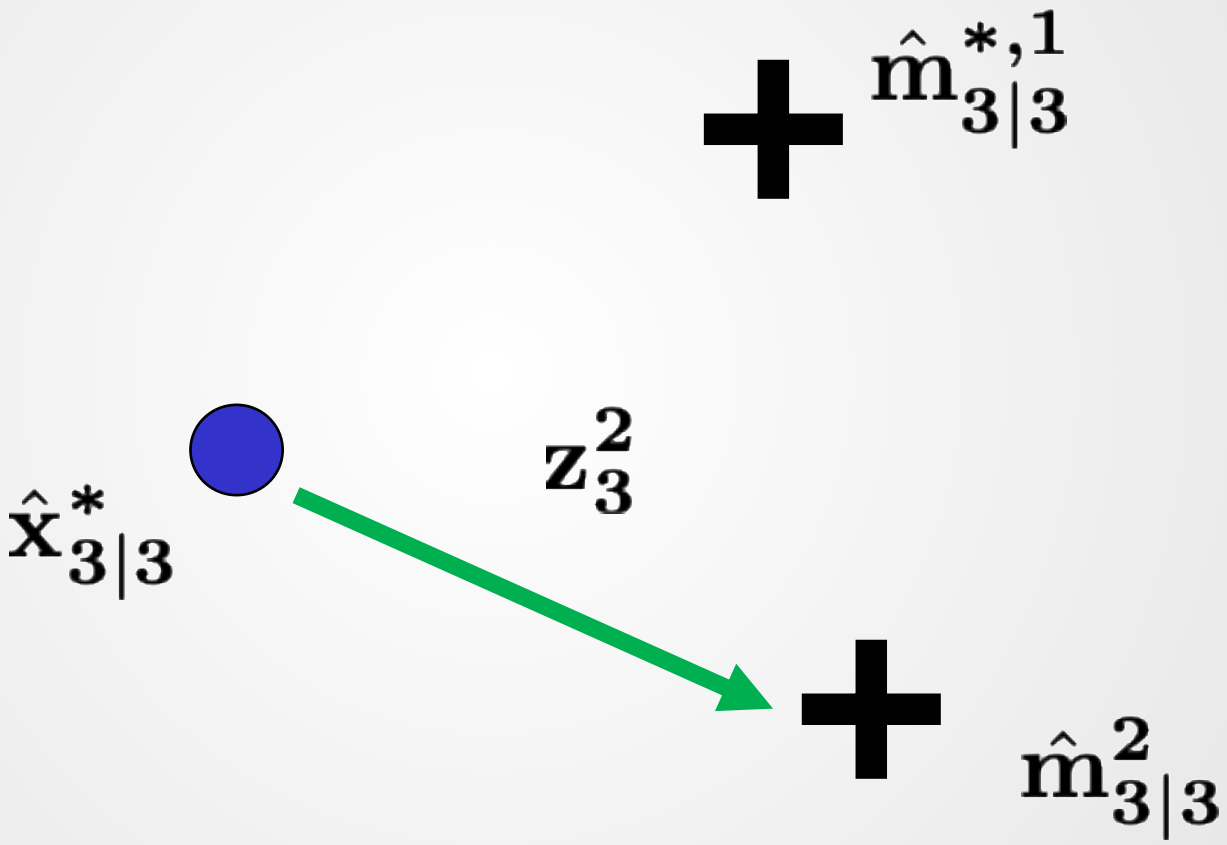
Timestep 3: Augmentation

- We use the same kind of augmentation operator again,

$$\hat{\mathbf{s}}_{3|3} = \mathbf{J}_3 \hat{\mathbf{s}}^*_{3|3} + \mathbf{K}_3 \mathbf{z}^2_3$$

- However, this needs to take account of the fact that we have a landmark already

Timestep 1: Augment with New Landmarks



Timestep 3: Augmentation

$$\hat{\mathbf{s}}_{3|3} = \begin{pmatrix} \hat{x}_{3|3} \\ \hat{\dot{x}}_{3|3} \\ \hat{m}^1_{3|3} \\ \hat{m}^2_{3|3} \end{pmatrix} = \begin{pmatrix} \hat{x}_{3|3} \\ \hat{\dot{x}}_{3|3} \\ \hat{m}^1_{3|3} \\ \hat{x}_{3|3} + z^1_3 \end{pmatrix}$$

Step 3: Augmentation Linear Operation

- **The matrices which will compute this are**

$$\mathbf{J_3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \mathbf{K_3} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Generalizing Timesteps and Multiple Observations

- *State Space Structure of the SLAM Problem*
- *Introduce the BeadSLAM Example*
- *Step-by-Step Example of the Kalman Filter SLAM System*
- **Generalizing Timesteps and Multiple Observations**
- *Analysing BeadSLAM*
- *Properties of Kalman Filter Based SLAM Systems*

Generalizing to Arbitrary Timesteps

- So far we've just worked through the first few timesteps with BeadSLAM
- We only considered one observation per timestep
- The equations we have so far readily generalize to multiple timesteps and multiple observations per timestep

General Time Update of the True System

- **The true system continues to evolve according to the equations,**

$$\mathbf{s}_{k+1} = \begin{bmatrix} \mathbf{x}_{k+1} \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{bmatrix} = \begin{bmatrix} \mathbf{F}_{k+1}\mathbf{x}_k + \mathbf{B}_{k+1}\mathbf{u}_{k+1} + \mathbf{v}_{k+1} \\ \mathbf{m}^1 \\ \vdots \\ \mathbf{m}^N \end{bmatrix}$$

Observing Multiple Landmarks

- Suppose we now M_k landmarks,

$$\mathbf{Z}_k = \left\{ \mathbf{z}_k^1, \ \mathbf{z}_k^2, \ \ldots, \ \mathbf{z}_k^{M_k} \right\}$$

- The identities of the landmarks are

$$I_k = \left\{ i_k^1, \ i_k^2, \ \ldots, \ i_k^{M_k} \right\}$$

Observing Multiple Landmarks

- The observation equation is now

$$\mathbf{Z}_k = \mathbf{H}_k^{I_k} \mathbf{x}_k + \mathbf{w}_k^{I_k}$$

Kalman Filter Prediction

- **The Kalman filter prediction is**

$$\hat{\mathbf{s}}_{k+1|k} = \mathbf{F}^{\mathbf{s}}_{k+1}\hat{\mathbf{s}}_{k|k} + \mathbf{B}^{\mathbf{s}}_{k+1}\mathbf{u}_{k+1}$$

$$\mathbf{P}^{s}_{k+1|k} = \mathbf{F}^{\mathbf{s}}_{k+1}\mathbf{P}^{s}_{k|k}\mathbf{F}^{\mathbf{s}}_{k+1}{}^{\top} + \mathbf{Q}^{s}_{k+1}$$

Step 1: Kalman Filter Prediction

- **The system equations simply expand to include all the landmarks we have,**

$$\mathbf{F}^{\mathbf{s}}_{k+1} = \begin{bmatrix} \mathbf{F}_{k+1} & 0 & 0 & \cdots & 0 \\ 0 & \mathbf{I} & 0 & \cdots & 0 \\ 0 & 0 & \mathbf{I} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \mathbf{I} \end{bmatrix}, \quad \mathbf{B}^{\mathbf{s}}_{k+1} = \begin{bmatrix} \mathbf{B}_k \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{Q}^{s}_{k+1} = \begin{bmatrix} \mathbf{Q}_{k+1} & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

Update and Augmentation

- The Kalman filter update equations can be repeatedly applied for each observation of each known landmark
- The augmentation process can be applied to add each landmark individually to the map

Analysing BeadSLAM

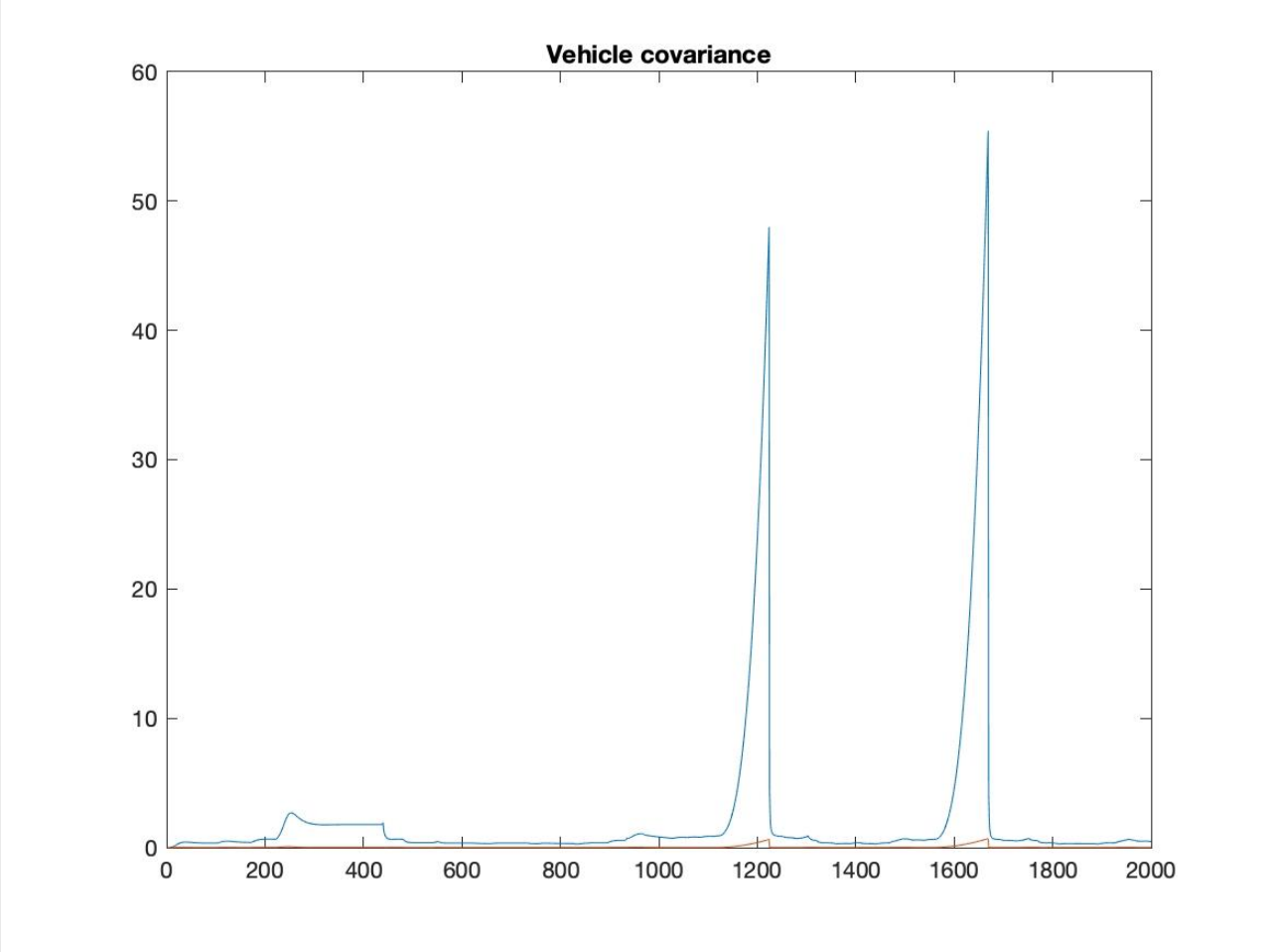
- *State Space Structure of the SLAM Problem*
- *Introduce the BeadSLAM Example*
- *Step-by-Step Example of the Kalman Filter SLAM System*
- *Generalizing Timesteps and Multiple Observations*
- **Analysing BeadSLAM**

Analysing BeadSLAM

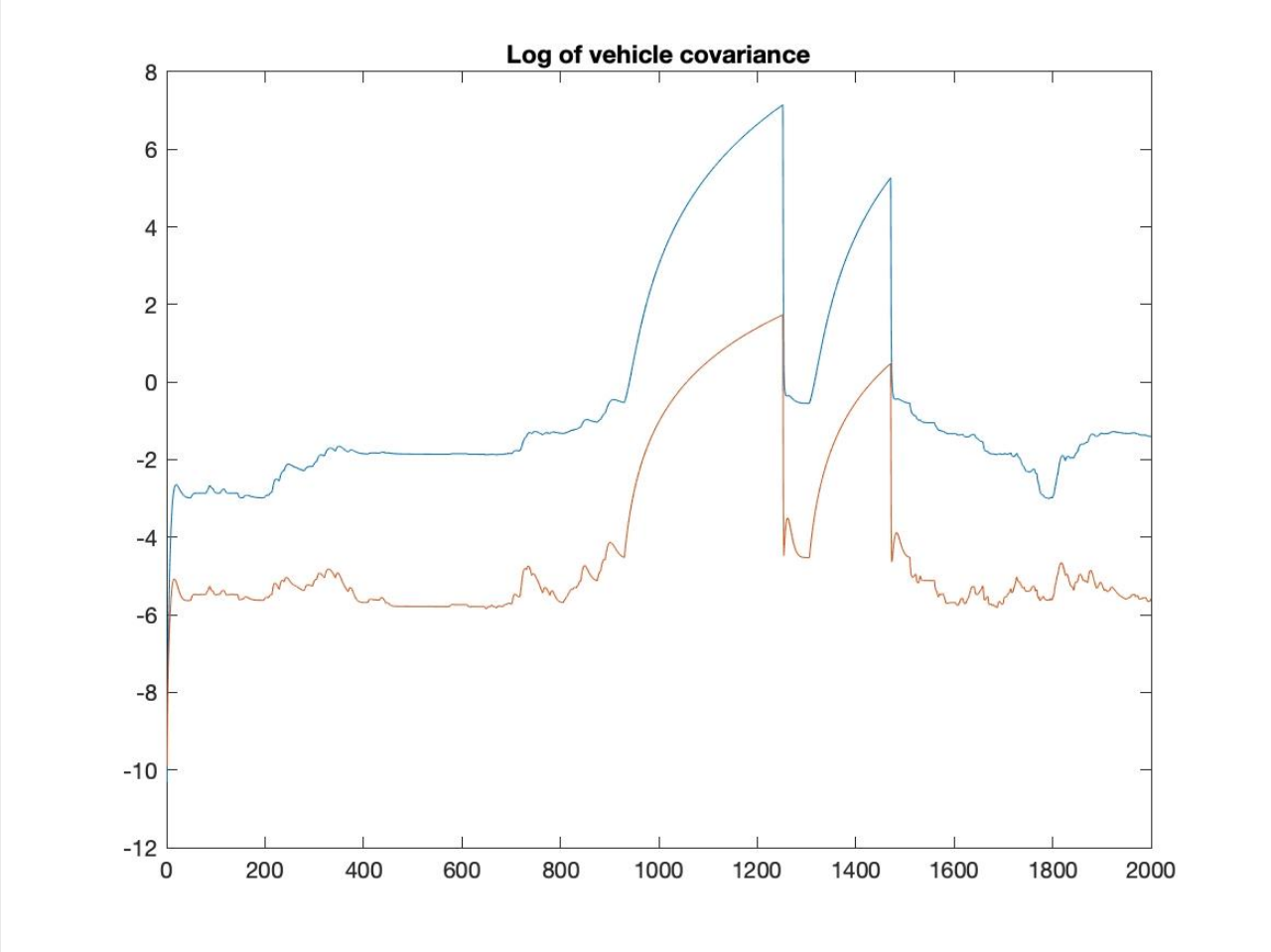
- We analyse BeadSLAM by looking at the covariance history only

- We look at the following:
  - The vehicle position covariance
  - The vehicle position / landmark correlations
  - The landmark position covariances

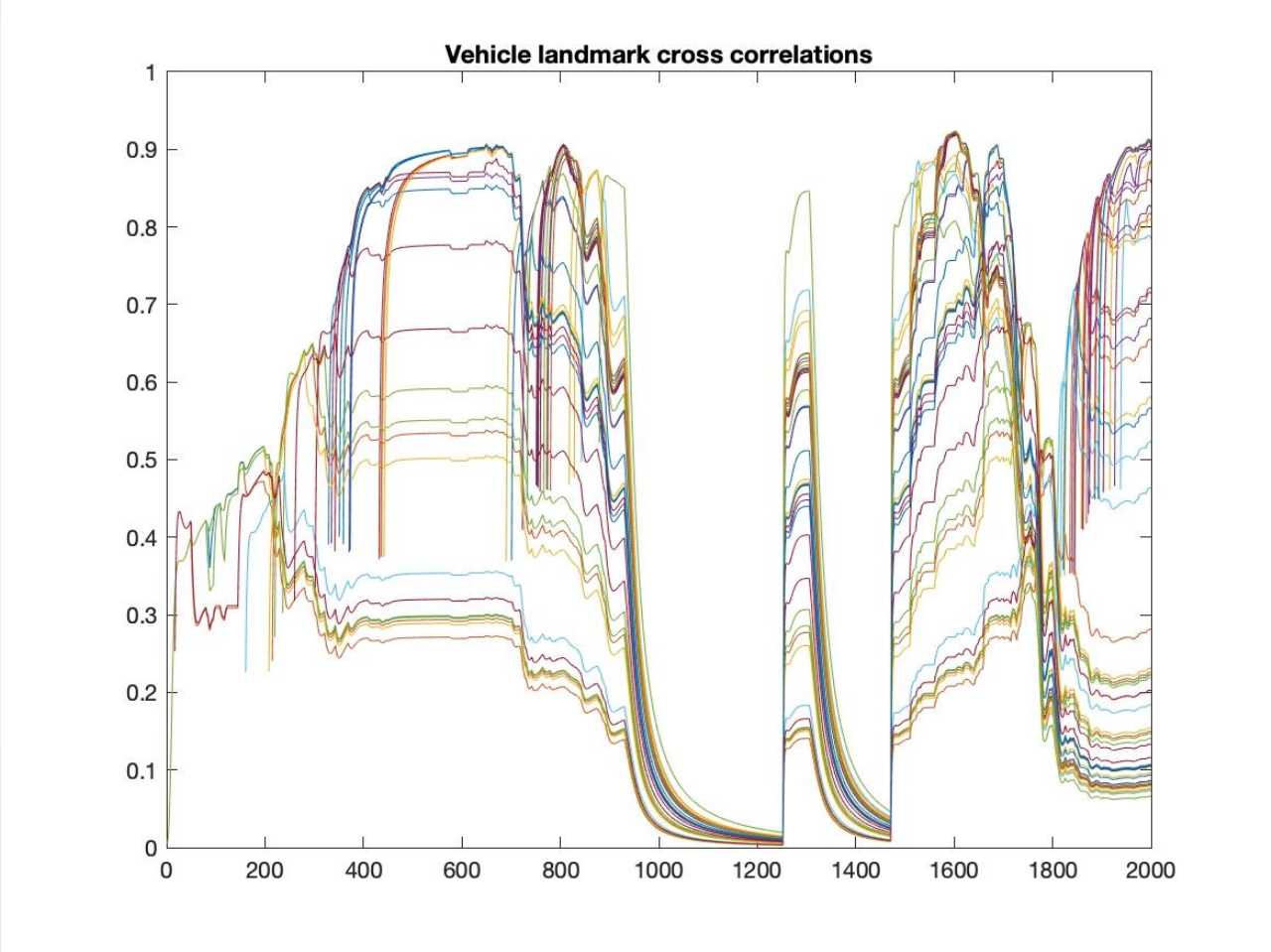
BeadSLAM Vehicle Covariance



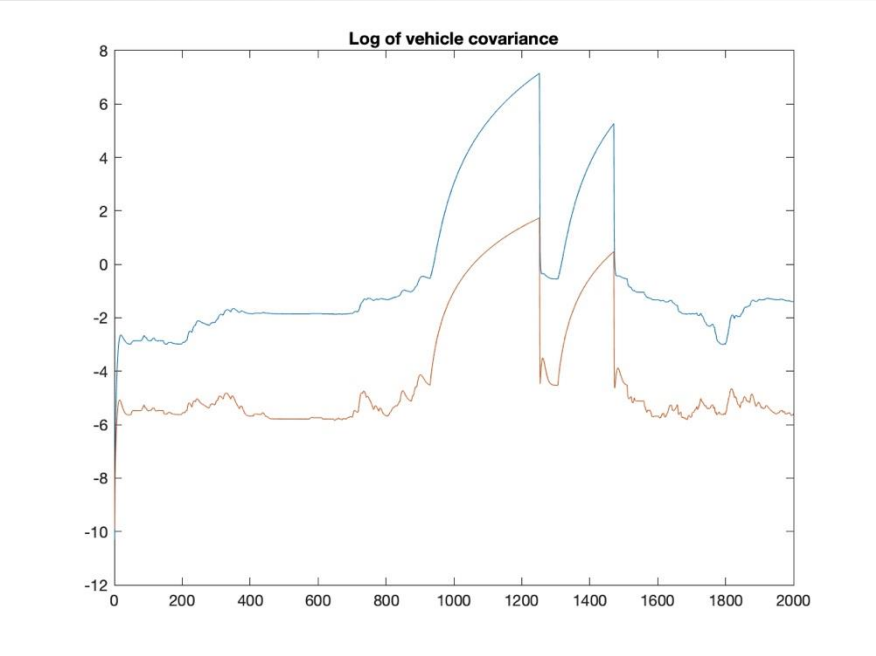
BeadSLAM Log Vehicle Covariance

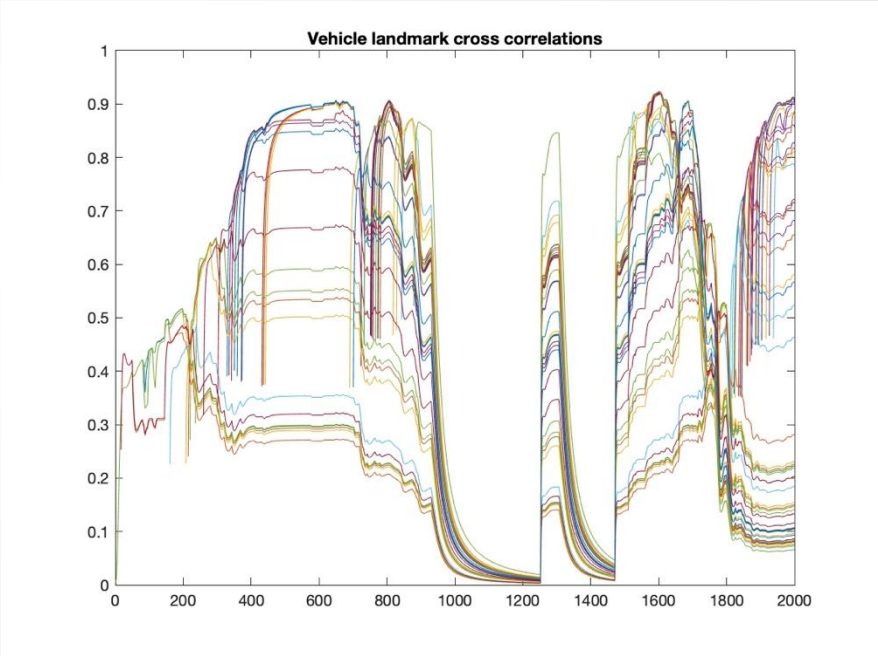


Landmark Vehicle Cross Correlation

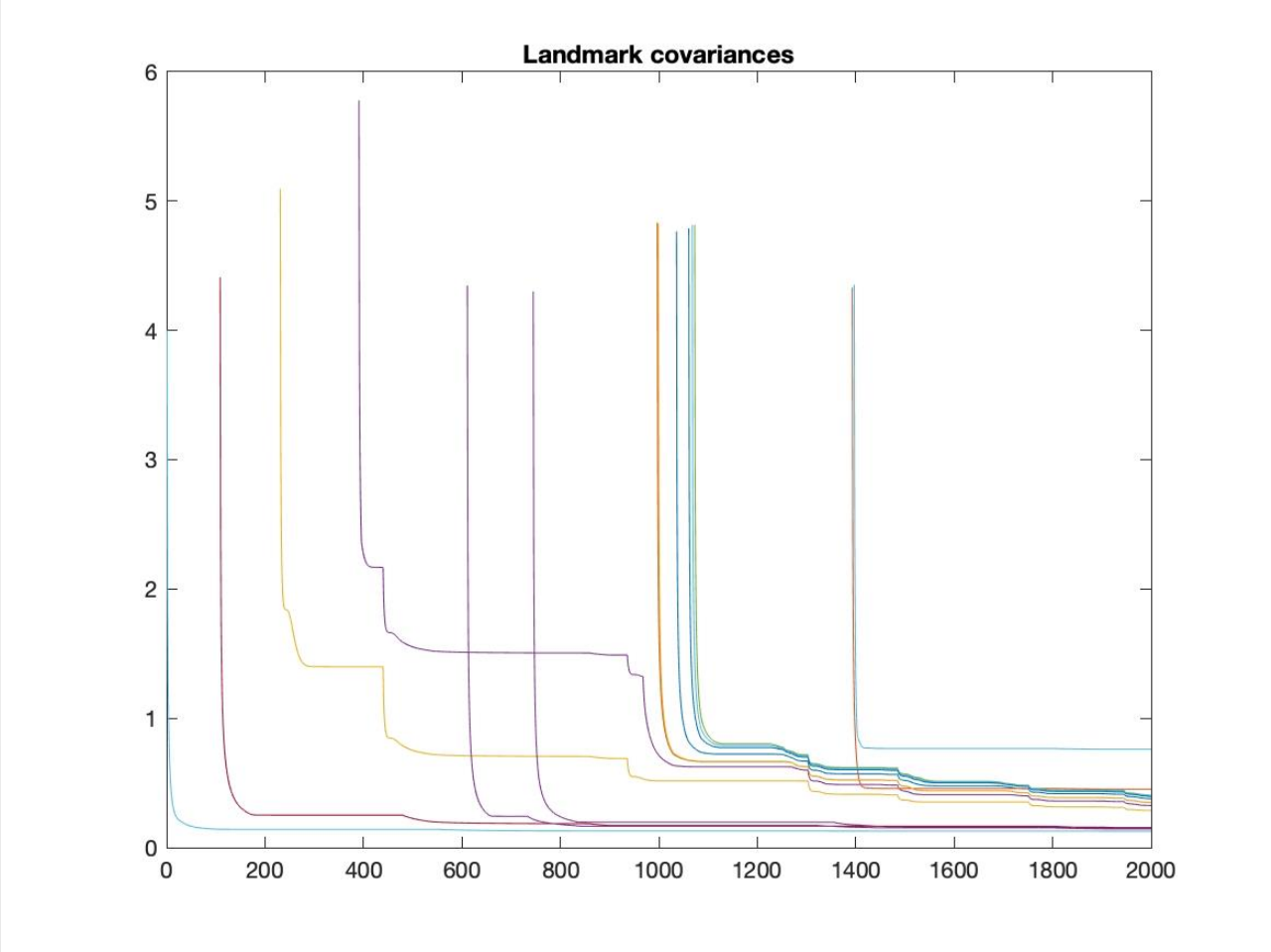


Landmark Vehicle Cross Correlation

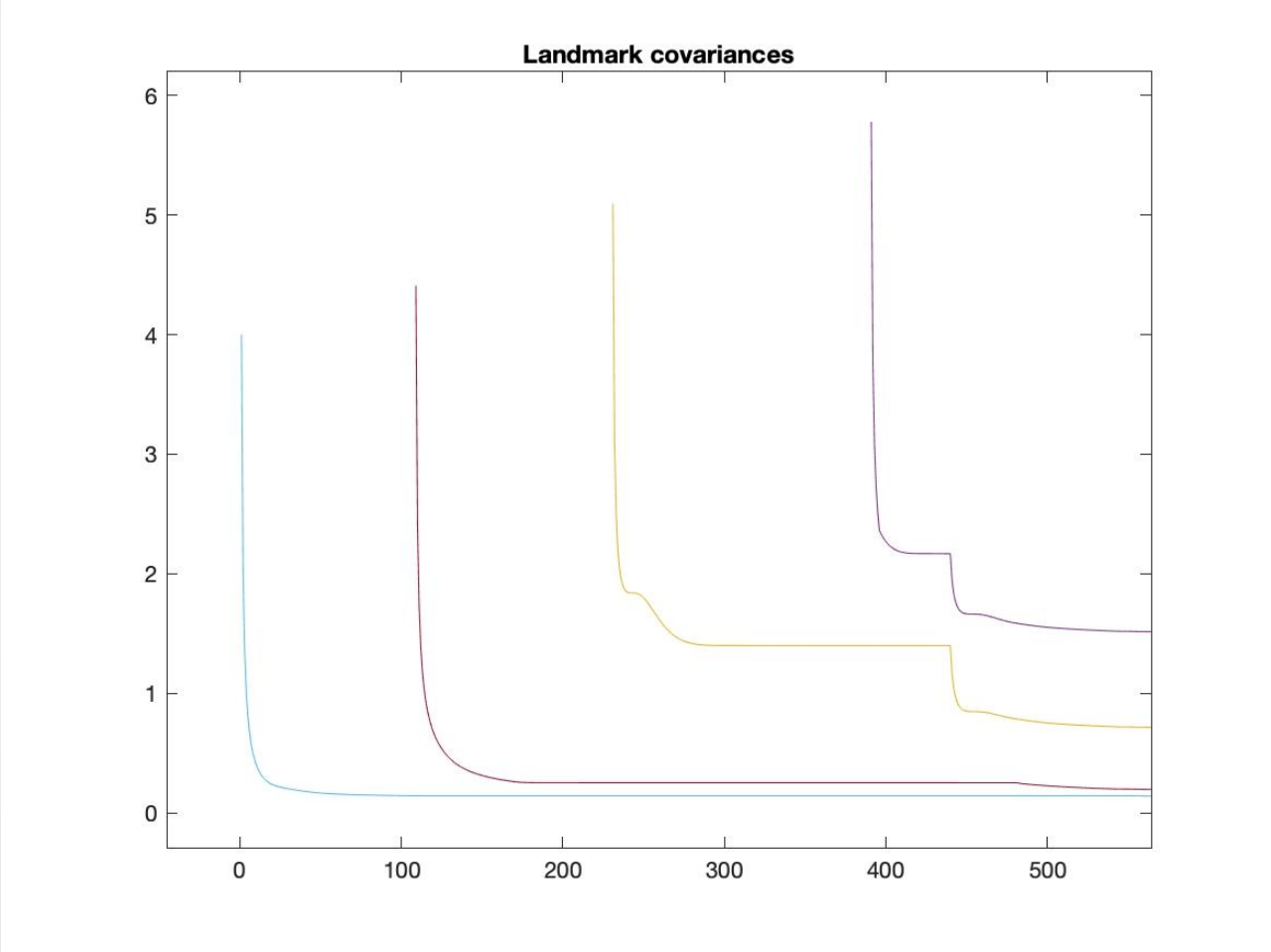




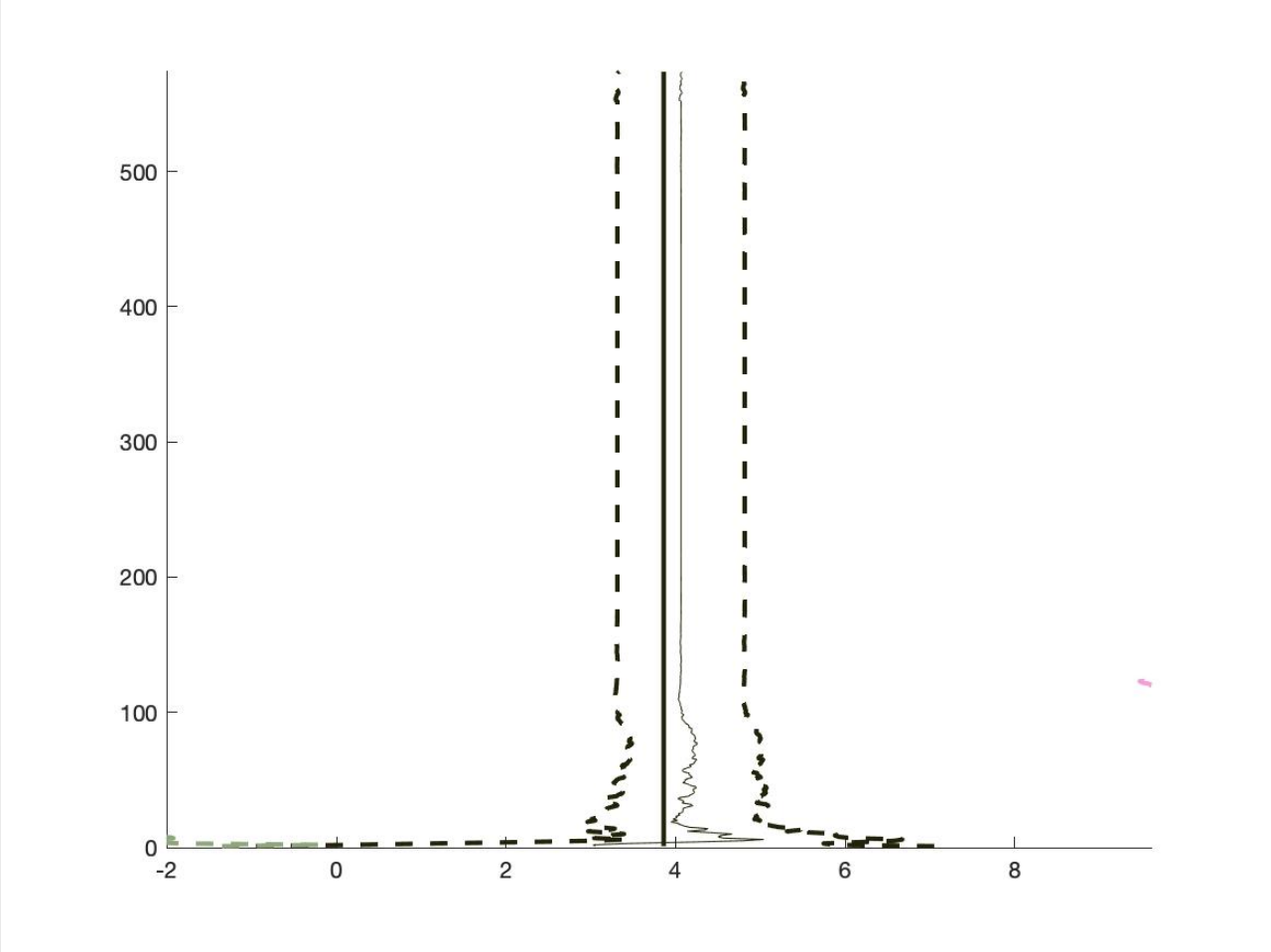
Landmark Covariances



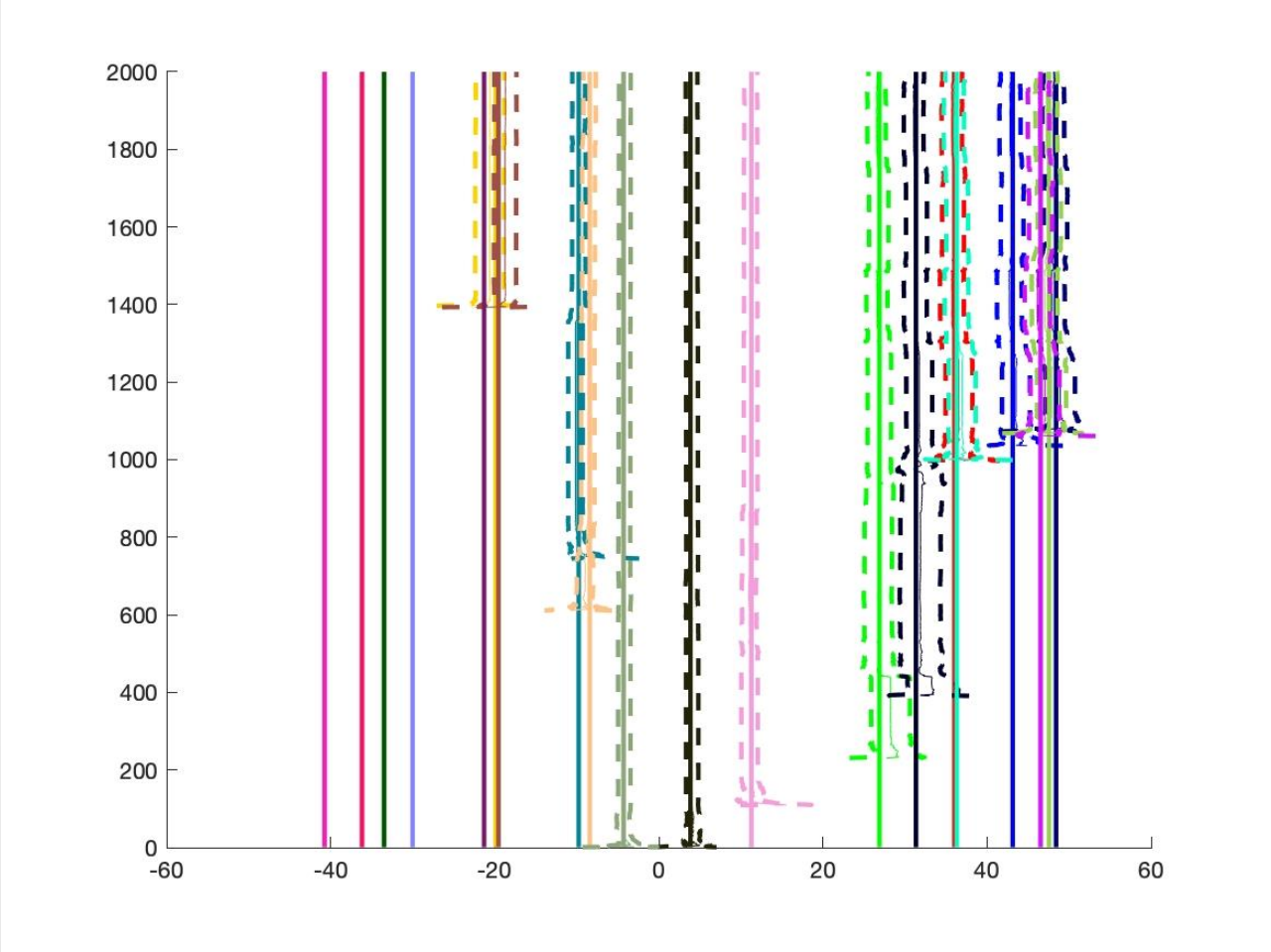
Landmark Covariance Behaviour



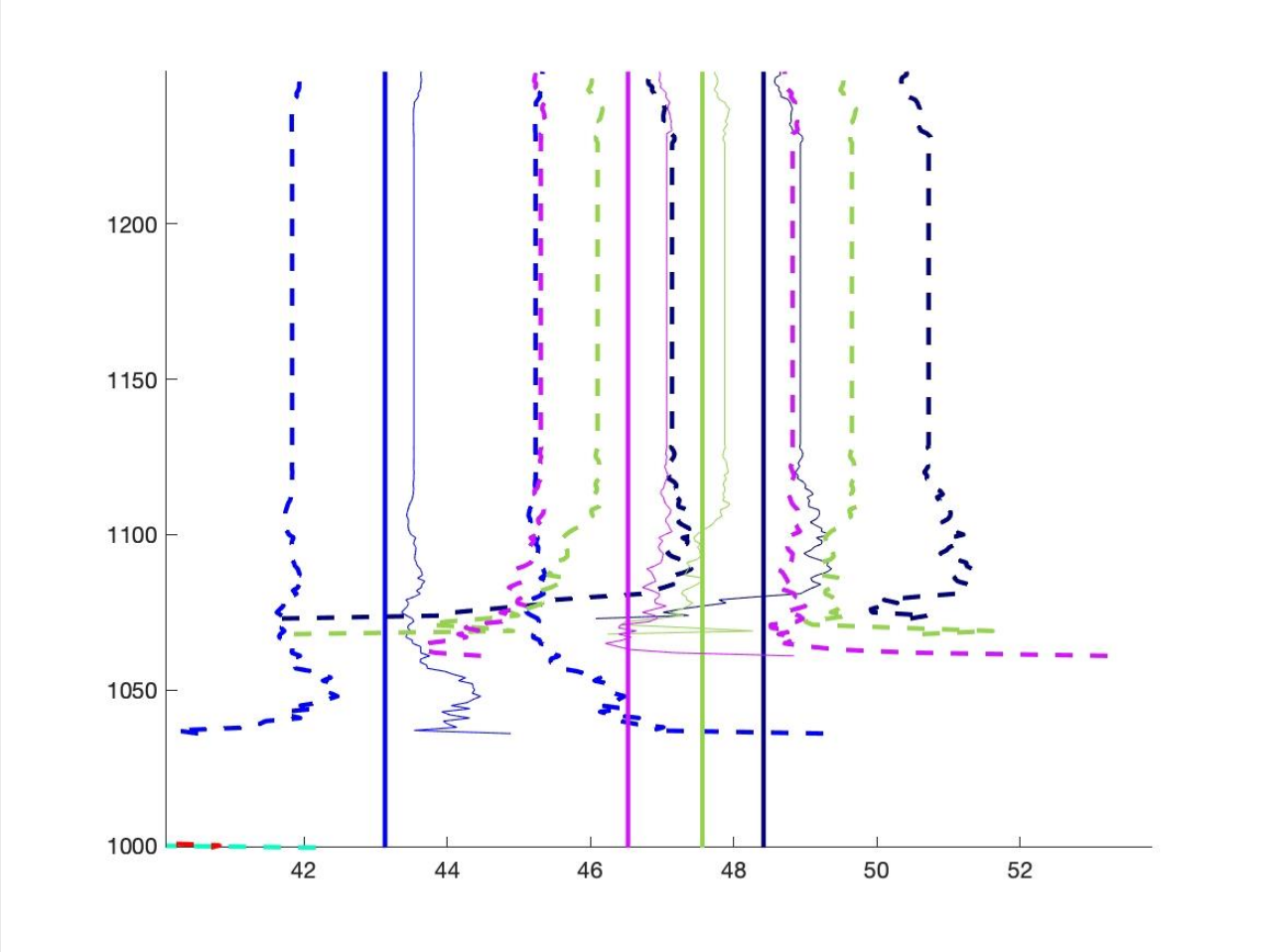
Waterfall Plot



Waterfall Plot



Waterfall Plot



Summary

- The linear Kalman filter can be used, with a linear system to develop a SLAM system
- We have demonstrated this for a small-scale problem for a small period of time
- How does it behave over a longer period of time?
- How does it behave in larger problems?
- In the next lecture, we will look at the scaling and convergence properties of linear Kalman Filter SLAM